SG No. 191 D_{6h}^{1} $P6/mmm$ [hexagonal]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: 1a, site symmetry: 6/mmm

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 1b, site symmetry: 6/mmm

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: 2c, site symmetry: -6m2

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,2,3,10,11,12,16,17,18,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[4,5,6,7,8,9,13,14,15,22,23,24]

Table 4: Wyckoff site: 2d, site symmetry: -6m2

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,2,3,10,11,12,16,17,18,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[4,5,6,7,8,9,13,14,15,22,23,24]

Table 5: Wyckoff site: 2e, site symmetry: 6mm

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,4,5,6,19,20,21,22,23,24]
2	$[0, 0, -z]$	[7,8,9,10,11,12,13,14,15,16,17,18]

Table 6: Wyckoff site: 3f, site symmetry: mmm

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,8,11,13,16,20,23]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,5,7,10,14,17,19,22]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,6,9,12,15,18,21,24]

Table 7: Wyckoff site: 3g, site symmetry: mmm

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,4,8,11,13,16,20,23]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,7,10,14,17,19,22]

continued ...

Table 7

No.	position	mapping
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3,6,9,12,15,18,21,24]

Table 8: Wyckoff site: 4h, site symmetry: 3m.

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,2,3,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[4,5,6,22,23,24]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9,13,14,15]
4	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[10,11,12,16,17,18]

Table 9: Wyckoff site: 6i, site symmetry: 2mm

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,4,20,23]
2	$\left[0, \frac{1}{2}, z\right]$	[2,5,19,22]
3	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[3,6,21,24]
4	$\left[0, \frac{1}{2}, -z\right]$	[7,10,14,17]
5	$\left[\frac{1}{2}, 0, -z\right]$	[8,11,13,16]
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,12,15,18]

Table 10: Wyckoff site: 6j, site symmetry: m2m

No.	position	mapping
1	$[x, 0, 0]$	[1,8,16,23]
2	$[0, x, 0]$	[2,7,17,22]
3	$[-x, -x, 0]$	[3,9,18,24]
4	$[-x, 0, 0]$	[4,11,13,20]
5	$[0, -x, 0]$	[5,10,14,19]
6	$[x, x, 0]$	[6,12,15,21]

Table 11: Wyckoff site: 6k, site symmetry: m2m

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,8,16,23]
2	$\left[0, x, \frac{1}{2}\right]$	[2,7,17,22]
3	$\left[-x, -x, \frac{1}{2}\right]$	[3,9,18,24]
4	$\left[-x, 0, \frac{1}{2}\right]$	[4,11,13,20]
5	$\left[0, -x, \frac{1}{2}\right]$	[5,10,14,19]
6	$\left[x, x, \frac{1}{2}\right]$	[6,12,15,21]

Table 12: Wyckoff site: 6l, site symmetry: mm2

No.	position	mapping
1	$[x,\ 2x,\ 0]$	[1,11,16,20]
2	$[-2x,\ -x,\ 0]$	[2,10,17,19]
3	$[x,\ -x,\ 0]$	[3,12,18,21]
4	$[-x,\ -2x,\ 0]$	[4,8,13,23]
5	$[2x,\ x,\ 0]$	[5,7,14,22]
6	$[-x,\ x,\ 0]$	[6,9,15,24]

Table 13: Wyckoff site: 6m, site symmetry: mm2

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{2}\right]$	[1,11,16,20]
2	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	[2,10,17,19]
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	[3,12,18,21]
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	[4,8,13,23]
5	$\left[2x,\ x,\ \frac{1}{2}\right]$	[5,7,14,22]
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	[6,9,15,24]

Table 14: Wyckoff site: 12n, site symmetry: ..m

No.	position	mapping
1	$[x,\ 0,\ z]$	[1,23]
2	$[0,\ x,\ z]$	[2,22]
3	$[-x,\ -x,\ z]$	[3,24]
4	$[-x,\ 0,\ z]$	[4,20]
5	$[0,\ -x,\ z]$	[5,19]
6	$[x,\ x,\ z]$	[6,21]
7	$[0,\ x,\ -z]$	[7,17]
8	$[x,\ 0,\ -z]$	[8,16]
9	$[-x,\ -x,\ -z]$	[9,18]
10	$[0,\ -x,\ -z]$	[10,14]
11	$[-x,\ 0,\ -z]$	[11,13]
12	$[x,\ x,\ -z]$	[12,15]

Table 15: Wyckoff site: 12o, site symmetry: .m.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,20]
2	$[-2x,\ -x,\ z]$	[2,19]
3	$[x,\ -x,\ z]$	[3,21]
4	$[-x,\ -2x,\ z]$	[4,23]
5	$[2x,\ x,\ z]$	[5,22]
6	$[-x,\ x,\ z]$	[6,24]
7	$[2x,\ x,\ -z]$	[7,14]

continued ...

Table 15

No.	position	mapping
8	$[-x, -2x, -z]$	[8,13]
9	$[-x, x, -z]$	[9,15]
10	$[-2x, -x, -z]$	[10,17]
11	$[x, 2x, -z]$	[11,16]
12	$[x, -x, -z]$	[12,18]

Table 16: Wyckoff site: 12p, site symmetry: m..

No.	position	mapping
1	$[x, y, 0]$	[1,16]
2	$[-y, x-y, 0]$	[2,17]
3	$[-x+y, -x, 0]$	[3,18]
4	$[-x, -y, 0]$	[4,13]
5	$[y, -x+y, 0]$	[5,14]
6	$[x-y, x, 0]$	[6,15]
7	$[y, x, 0]$	[7,22]
8	$[x-y, -y, 0]$	[8,23]
9	$[-x, -x+y, 0]$	[9,24]
10	$[-y, -x, 0]$	[10,19]
11	$[-x+y, y, 0]$	[11,20]
12	$[x, x-y, 0]$	[12,21]

Table 17: Wyckoff site: 12q, site symmetry: m..

No.	position	mapping
1	$\left[x, y, \frac{1}{2}\right]$	[1,16]
2	$\left[-y, x-y, \frac{1}{2}\right]$	[2,17]
3	$\left[-x+y, -x, \frac{1}{2}\right]$	[3,18]
4	$\left[-x, -y, \frac{1}{2}\right]$	[4,13]
5	$\left[y, -x+y, \frac{1}{2}\right]$	[5,14]
6	$\left[x-y, x, \frac{1}{2}\right]$	[6,15]
7	$\left[y, x, \frac{1}{2}\right]$	[7,22]
8	$\left[x-y, -y, \frac{1}{2}\right]$	[8,23]
9	$\left[-x, -x+y, \frac{1}{2}\right]$	[9,24]
10	$\left[-y, -x, \frac{1}{2}\right]$	[10,19]
11	$\left[-x+y, y, \frac{1}{2}\right]$	[11,20]
12	$\left[x, x-y, \frac{1}{2}\right]$	[12,21]

Table 18: Wyckoff site: 24r, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-y, x-y, z]$	[2]

continued ...

Table 18

No.	position	mapping
3	$[-x+y,\ -x,\ z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[y,\ -x+y,\ z]$	[5]
6	$[x-y,\ x,\ z]$	[6]
7	$[y,\ x,\ -z]$	[7]
8	$[x-y,\ -y,\ -z]$	[8]
9	$[-x,\ -x+y,\ -z]$	[9]
10	$[-y,\ -x,\ -z]$	[10]
11	$[-x+y,\ y,\ -z]$	[11]
12	$[x,\ x-y,\ -z]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$[y,\ -x+y,\ -z]$	[14]
15	$[x-y,\ x,\ -z]$	[15]
16	$[x,\ y,\ -z]$	[16]
17	$[-y,\ x-y,\ -z]$	[17]
18	$[-x+y,\ -x,\ -z]$	[18]
19	$[-y,\ -x,\ z]$	[19]
20	$[-x+y,\ y,\ z]$	[20]
21	$[x,\ x-y,\ z]$	[21]
22	$[y,\ x,\ z]$	[22]
23	$[x-y,\ -y,\ z]$	[23]
24	$[-x,\ -x+y,\ z]$	[24]